

tar & Compression Cheat Sheet

Bundle files and folders into one archive, squeeze them smaller, and unpack them again.

Using it

```
tar -czvf backup.tar.gz project/  ← create a gzipped archive of project/
tar -xzvf backup.tar.gz           ← extract it into the current folder
tar -tzvf backup.tar.gz           ← list what's inside without extracting
```

Read the letters as words: **create** / **extract** / **list**, **zipped**, **verbose**, **file**. Keep **f** last (the archive name follows it).

Option	What it does	Option	What it does
-c	Create a new archive	-z	gzip compression (.tar.gz , .tgz)
-x	Extract files	-j	bzip2 compression (.tar.bz2)
-t	List the contents	-J	xz compression (.tar.xz , smallest)
-f name	The archive file to use	-C dir	Extract into dir (or create from dir)
-v	Show each file as it goes	--exclude='*.log'	Leave out matching files
-p	Keep exact permissions	--strip-components=1	Drop the top folder when extracting

When extracting, tar spots the compression itself, so **tar -xf file.tar.xz** works without **-J**.

Recognizing archive types

Ending	What it is	Unpack with
.tar	Bundle only, not compressed	tar -xf file.tar
.tar.gz .tgz	tar bundle, gzip-compressed (the most common)	tar -xzf file.tar.gz
.tar.bz2	tar bundle, bzip2-compressed	tar -xjf file.tar.bz2
.tar.xz	tar bundle, xz-compressed	tar -xJf file.tar.xz
.gz	One single file, gzip-compressed (no bundle)	gunzip file.gz
.zip	Bundle and compression in one (Windows-friendly)	unzip file.zip

gzip and zip

Command	Does	Command	Does
gzip file	Compresses to file.gz (removes file)	zip -r out.zip dir/	Zips a folder
gzip -k file	Same, but keeps the original	unzip out.zip	Unzips here
gunzip file.gz	Uncompresses (same as gzip -d)	unzip -l out.zip	Lists the contents
zcat file.gz	Prints it without uncompressing	unzip x.zip -d dir	Unzips into dir

Reading the output of tar -tvf

```
$ tar -tzvf backup.tar.gz
drwxr-xr-x mark/devs      0 2026-03-04 09:12 project/  ← d = folder
-rw-r--r-- mark/devs    1834 2026-03-04 09:10 project/README.md
-rwxr-xr-x mark/devs     612 2026-03-02 17:45 project/run.sh
|           |           |   |
perms      owner/group  size date+time      path to use when extracting
```

tar strips the leading /, so files land under the folder you extract in. Copy a path as listed to extract one file.

Handy recipes

Command	Does
tar -xzf backup.tar.gz project/README.md	Extracts just one file (path as shown by -t)
tar -xzf app.tar.gz -C /opt	Extracts into /opt instead of here
tar -czf site.tgz --exclude='*.log' site/	Backs up a folder, skipping log files
tar -czf etc-\$(date +%F).tar.gz /etc	Dated backup, e.g. etc-2026-03-04.tar.gz
tar -cf - dir ssh host 'tar -xf - -C /dst'	Copies a folder to another server in one go

Remember: list before you extract (**tar -tf**). Archives with no top folder spill files all over the current folder.